

Fidelity of Implementation of Research-Based Instructional Strategies (RBIS) in Engineering Science Courses

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Abstract

Background Increasing attention is being paid to improvement in undergraduate science, technology, engineering, and mathematics (STEM) education through increased adoption of research-based instructional strategies (RBIS), but high-quality measures of faculty instructional practice do not exist to monitor progress.

Purpose/Hypothesis The measure of how well an implemented intervention follows the original is called *fidelity of implementation*. This theory was used to address the research questions: What is the fidelity of implementation of selected RBIS in engineering science courses? That is, how closely does engineering science classroom practice reflect the intentions of the original developers? Do the critical components that characterize an RBIS discriminate between engineering science faculty members who claimed use of the RBIS and those who did not?

Design/Method A survey of 387 U.S. faculty teaching engineering science courses (e.g., statics, circuits, thermodynamics) included questions about class time spent on 16 critical components and use of 11 corresponding RBIS. Fidelity was quantified as the percentage of RBIS users who also spent time on corresponding critical components. Discrimination between users and nonusers was tested using chi square.

Results Overall fidelity of the 11 RBIS ranged from 11% to 80% of users spending time on all required components. Fidelity was highest for RBIS with one required component: case-based teaching, just-in-time teaching, and inquiry learning. Thirteen of 16 critical components discriminated between users and nonusers for all RBIS to which they were mapped.

Conclusions Results were consistent with initial mapping of critical components to RBIS. Fidelity of implementation is a potentially useful framework for future work in STEM undergraduate education.

Keywords engineering science; fidelity of implementation; research-based instructional strategies (RBIS)

Introduction

Increasingly, reports from high-profile organizations in the United States are calling for widespread improvements to engineering and science, technology, engineering, and mathematics

(STEM) undergraduate education (e.g., American Society for Engineering Education [ASEE], 2009; Association of American Universities [AAU], 2011; Committee on Science Engineering and Public Policy, 2006, 2010; National Science Board, 2010). The most recent of these (ASEE, 2012; National Research Council [NRC], 2012; President's Council of Advisors on Science and Technology, 2012) draw attention to the complex processes of encouraging a majority of faculty to use research-based instructional strategies (RBIS) in their engineering, math, and science courses. The focus is shifting from research and development of RBIS such as active learning (Prince, 2004), cooperative learning (Prince, 2004), and problem-based learning (Prince & Felder, 2006) to include emphasis on faculty and institutional change processes. The recent DBER report, *Discipline-Based Educational Research: Understanding and Improving Learning in Undergraduate Science and Engineering* (NRC, 2012), specifically calls for more research in this area.

In order to measure progress toward goals of widespread use of RBIS, a number of challenges must be addressed. Among these is developing complete and consistent descriptions of each RBIS, so that reports and measures of usage and efficacy are both accurate and precise. An apparent lack of this type of consistency would explain in part why there is often disagreement concerning the efficacy of specific RBIS, including those with long histories of development and assessment.

Research publications often target other researchers, rather than practitioners, so detailed descriptions of methodology focus on providing enough information to replicate the study itself – not the intervention. Rarely is sufficient detail provided to faithfully implement the instructional strategy. Publications that focus on practitioners tend to emphasize raising awareness and communicating effectiveness (Borrego, Froyd, & Hall, 2010), sometimes at the expense of omitting important details about how to implement the instructional strategies. This level of detail is important, given the complexity and diversity of engineering students and learning environments. Specific instructional strategies are likely to work in *certain* settings under *certain* conditions – not all of them. In order to understand and communicate these conditions, it is important to have complete and accurate descriptions and measures of how and when an instructional strategy is being implemented.

In the K–12 educational system, careful documentation of implementation has become necessary in local, state, and national implementation of programs, creating a strong need for accurate description of each program site (Century, Rudnick, & Freeman, 2010). The measure of how well the implemented intervention follows the original is called *fidelity of implementation* (O'Donnell, 2008). The purpose of this survey research is to explore how fidelity of implementation theory from K–12 education applies to engineering undergraduate education. Specifically, we address the following research questions:

What is the fidelity of implementation of selected RBIS in engineering science courses? That is, how closely does engineering science classroom practice reflect the intentions of the original developers?

Do the critical components that characterize a RBIS discriminate between engineering science faculty who claimed use of the RBIS and those who did not?

The results of this survey will inform change efforts in engineering education and undergraduate education in other STEM disciplines. This study mirrors similar investigations in physics (Henderson & Dancy, 2009) and geosciences (Macdonald, Manduca, Mogk, & Tewksbury, 2005), and many of the RBIS we are studying were developed in

other STEM fields. We chose to focus this study on engineering science courses for several reasons. Engineering sciences (e.g., statics, circuits, and thermodynamics) are the core of engineering. They distinguish engineering from other disciplines and are also the gateway courses that must be passed by all students to continue in engineering. In many ways, these courses are the equivalent of introductory science courses in which these RBIS were originally developed. Along with junior-level courses specific to the major, they appear to be the most resistant to pedagogical change. This resistance is evidenced, for example, by the bookending of active learning design experiences now typical in the freshman and senior years (Froyd, 2005).

Literature Review

Our literature review comprises two parts. First, we review the literature on fidelity of implementation (FOI). We then describe the specific research-based instructional strategies (RBIS) we studied, including selection criteria and evidence of efficacy.

Fidelity of Implementation

According to Rogers (2003), early dissemination studies were based on the implicit assumption that all adoption was “an exact copying or imitation of how the innovation had been used previously in a different setting” (p. 174). More recently, it has become clear that studies must also consider the extent to which the innovation changes during the diffusion process; therefore, researchers have begun considering fidelity of implementation. Fidelity of implementation is broadly defined as the measure of how well an implemented intervention follows the original (O'Donnell, 2008). While there is limited research in engineering education specifically investigating fidelity of implementation, fields such as mental health, program evaluation, education, and human services have been conducting such investigations for several decades (Mowbray et al., 2003).

A high level of fidelity has obvious benefits for being able to measure and trace diffusion of an innovation, e.g. a specific teaching strategy. However, adopters (particularly engineering faculty) want or need to modify the innovation in order for it to work in their local environment. Some level of adaptation is desirable because it promotes adoption and may also improve upon the innovation (Rogers, 2003). Work in fidelity of implementation is beginning to consider these issues holistically, but debate continues about the extent to which an innovation should be allowed to change (Blakey, 1987; Century et al., 2010; Emshoff et al., 1987; Hall & Loucks, 1978; Mowbray, Holter, Teague, & Bybee, 2003; O'Donnell, 2008).

There are typically two types of fidelity studies: efficacy and effectiveness. Aspects of both are relevant to our study of undergraduate engineering education. An efficacy study investigates whether the intervention can achieve the set objectives under ideal circumstances. According to O'Donnell (2008), “Efficacy study's examination of fidelity focuses on whether a program is implemented at all . . . and to what degree . . . and it uses the answers to these questions to improve the program” (p. 41). Efficacy studies typically focus on the development stage and help developers to critically analyze the needed components for the innovation to succeed or fail. Many engineering education researchers, particularly those developing curricular materials or instructional strategies, conduct efficacy-type studies. When they expand their work to include other student populations at other institutions, for example, they are trying to better understand the exact conditions under which their interventions improve student learning or motivation.

An effectiveness study, in contrast, investigates the effects of an innovation when implemented by regular users in actual practice. As described by O'Donnell, "Effectiveness studies seem more interested in interpreting evidence of the program for generalizability . . . and observing the implementation of the program in the field" (p. 42). The focus shifts from the level of fidelity needed for the intervention to be successful (efficacy studies) to the changes that users make in practice and the impact of these changes on the ability of the intervention to achieve its goals (effectiveness studies). Ours is an effectiveness study because it investigates how RBIS are being used in by engineering science faculty. Fidelity is a relatively new concept in undergraduate STEM education, and previous efficacy studies on RBIS did not contain a fidelity element. Therefore, this article includes more discussion of a number of elements typically considered in efficacy studies, such as identification of critical components. For further discussion and examples of efficacy and effectiveness studies as well as a general review of fidelity of implementation research, see O'Donnell (2008).

A key element of fidelity is the components that characterize the intervention. Century et al. (2010) called these the *critical components*, defined as the "essential features that must be measured to determine whether a program is present or not" (p. 201). Similar to engineering specifications used in the design process, critical components help indicate whether the intervention does what was intended. Critical components are similar to the *components* in the concerns-based adoption model's (CBAM's) innovation configurations. Mowbray, Holter, Teague, and Bybee (2003) refer to them as *fidelity criteria*, Carroll et al. (2007) call them *essential components*, and Bickman et al. (2009) label them *fidelity measures*. Despite the variations in terminology, each of these perspectives focuses on identifying the components that are necessary for fidelity. In this study, we will refer to them as *critical components*. Efficacy studies develop the critical components, while effectiveness studies investigate whether the critical components are present or if they have been changed. In order to conduct an effectiveness study, we must first determine the critical components of the RBIS of interest.

The actual steps to identifying critical components are laid out in CBAM's innovation configuration literature (Hall & Loucks, 1978). Developing a configuration begins with a literature review of the innovation, followed by an iterative process that includes user interviews and observations. Next, the component checklist is constructed, which includes all of the components and their variations. The final step is locating dominant patterns by categorizing individual users (from earlier interviews) by the variations they use. As a result, the list may be revised. The level of fidelity in a particular setting can then be quantified by dividing the number of components implemented by the total number of components. For examples of fidelity research that have used CBAM, see Anderson (1997), Emshoff et al. (1987), and Mills and Ragan (2000).

The process of establishing an innovation configuration is a significant undertaking; it is usually performed by the developer of the innovation, rarely (if ever) for more than one intervention at a time. For the purposes of this study, in which we are comparing 11 RBIS, we have abbreviated the process to include an extensive literature search and consensus among a panel of engineering and physics education experts. We note, however, that systematically identifying critical components through innovation configurations or other means is an important direction for future work, particularly among the developers of new RBIS.

Mowbray et al. (2003) reviewed the literature on fidelity of implementation to identify common steps used to establish, measure, and validate the fidelity criteria of an innovation; these steps are outlined in Table 1. The first step is to establish critical components.

Table 1 Mowbray et al.’s (2003) Steps to Evaluate Fidelity of Implementation (italics indicate methods used in this study)

Steps	Methods to accomplish each
1. Establish critical components	Established program’s model (listing components) <i>Literature/Expert opinion</i> Qualitative research methods (e.g., user interviews)
2. Measure fidelity	Researcher/Expert ratings from observations or documentation <i>User surveys</i> User interviews
3. Validate fidelity criteria	<i>Inter-rater reliability</i> <i>Comparison of actual to expected results</i> Comparison across known groups/programs Comparison of fidelity to efficacy measures <i>Convergent validity of multiple sources</i>

The three most common ways of establishing critical components are using an established program model that already lists critical components, using the expert opinion of those who have published on the innovation, and using qualitative research methods such as interviews with users. These are abbreviated steps that reflect CBAM’s innovation configuration. For this study, we relied on the literature, augmented by the expertise of our research team, which includes authors of the most frequently cited articles related to RBIS in engineering.

Once critical components have been identified, the next steps are to actually measure fidelity and establish validity (Mowbray et al., 2003). The actual measurement of fidelity can be accomplished through researcher or expert ratings of observations or other implementation documentation, user surveys, or user interviews. In this study, we surveyed engineering science faculty about their classroom activities and RBIS use. The validity and reliability of the criteria can be established utilizing one or more of the following methods: ensuring inter-rater reliability, confirming consistency of the internal structure between expected and actual results, ensuring convergent validity by using multiple sources, using known groups to compare across programs, and comparing the observed fidelity measures to the expected outcomes of the user (e.g., student learning gains). When establishing the critical components in this study, we ensured inter-rater reliability by discussing and refining the set of critical components until all five team members agreed. Additionally, we surveyed faculty from multiple universities and engineering disciplines (aerospace, chemical, civil, computer, electrical, and mechanical engineering) to compare fidelity across settings. Throughout the Results section and in the discussion that follows, we compare the actual results to the expected results and consider implications for fidelity criteria.

Research-Based Instructional Strategies

We used the following criteria for selection of RBIS to be included in this study: documented use in engineering settings at more than one institution and demonstrated positive influence on student learning in engineering or STEM.

We acknowledge that the evidence base for some of the most widely recommended undergraduate STEM instructional strategies does not meet the highest standards identified by hierarchies of evidence used in medicine and K–12 education (e.g., U.S. Department of

Education, 2007). For example, the What Works Clearinghouse (2011) of the U.S. Department of Education's Institute for Education Sciences excludes from its reviews any study that does not include a comparison or control group demonstrated to be equivalent to the treatment group. Given the autonomy of undergraduate faculty to develop, select, and adapt their own instructional strategies, this level of standardization may not be realistic.

The DBER report, focusing on undergraduate STEM education, concludes that "research-based instructional strategies are more effective than traditional lecture in improving conceptual knowledge and attitudes about learning" (NRC, 2012, p. 3). The report concludes, "Overall the committee has characterized the strength of the evidence on making lecture more interactive as strong" – the highest level in their hierarchy of evidence – "because of the high degree to which the findings converge" (p. 122). Specific strategies discussed in this section of the report overlap significantly with those in this study: just-in-time teaching, concept tests, think-pair-share, collaborative learning, cooperative learning, and use of peer instruction (with the caveat that these strategies are used appropriately).

Additionally, we note that a number of high-profile undergraduate STEM education efforts are underway (e.g., AAU, 2011; Bay View Alliance, n.d.; Business-Higher Education Forum, n.d.) and will not wait for more rigorous evidence to accumulate. Studies and methods to understand and measure use of instructional strategies must develop in parallel. This study addresses the latter by introducing the concept of fidelity of implementation to STEM undergraduate education so that implementation can be understood to evolve as the evidence base is strengthened.

Table 2 summarizes the RBIS explored in this study, and the following text briefly describes each RBIS and reviews evidence of their efficacy. Note that although there is significant overlap in the way that these terms can be used, the more detailed descriptions in the text below seek to clearly categorize a particular instructional activity as a single RBIS.

Just-in-time teaching Just-in-time teaching (JiTT) combines Web-based technology with active learning methods in the classroom (Modesitt, Maxim, & Akingbehin, 1999; Novak, Patterson, Gavrin, & Christian, 1999; Rozycki, 1999). Students individually complete Web-based assignments a few hours before class, then the instructor reads through their answers before class and adjusts the lessons accordingly ("just in time"). Novak et al. (1999) assessed JiTT for its impact on cognitive outcomes, student attrition, and student attitudes in physics. Student learning was assessed using the Force Concept Inventory, which showed normalized student gains between 35% and 40%. This gain is similar to that found for other interactive-engagement teaching methods (Hake, 1998) and is significantly better than the average normalized gains found in traditionally taught physics courses. Novak et al. (1999) also reported that student attrition was 40% less with JiTT than with traditionally taught courses. Cashman and Eschenbach (2003) also found that JiTT increased student engagement and confidence when implemented in their Introduction to Environmental Engineering course.

Case-based teaching Case-based teaching requires students to analyze case studies of historical or hypothetical situations that involve solving problems or making decisions (Prince & Felder, 2007). Studies have found that compared with conventional strategies, case-based teaching significantly improves retention (Fasko, 2003), reasoning and problem-solving skills (Fasko, 2003; Levin, 1997), higher-order skills on Bloom's taxonomy (Gabel, 1999), the ability to make objective judgments (Dinan, 2002), the ability to identify relevant issues and recognize multiple perspectives, and awareness of ethical issues (Lundeberg, Levin, & Harrington, 1999). A meta-analysis by Lundeberg and Yadav (2006a, 2006b) concluded that

Table 2 Research-Based Instructional Strategies (RBIS) and Descriptions Supplied to Participants as They Completed the Survey

RBIS	Brief description
Just-in-time teaching	Asking students to individually complete homework assignments a few hours before class, reading through their answers before class, and adjusting the lessons accordingly
Case-based teaching	Asking students to analyze case studies of historical or hypothetical situations that involve solving problems and/or making decisions
Service learning	Intentionally integrating community service experiences into academic courses to enhance the learning of the core content and to give students broader learning opportunities about themselves and society at large
Think-aloud-paired problem solving	Forming pairs in which one student works through a problem while the other questions the problem solver in an attempt to get them to clarify their thinking
Inquiry learning	Introducing a lesson by presenting students with questions, problems, or a set of observations and using this to drive the desired learning
Peer instruction	A specific way of using concept tests in which the instructor poses the conceptual question in class and then shares the distribution of responses with the class (possibly using a classroom response system or “clickers”). Students form pairs, discuss their answers, and then vote again
Concept tests	Asking multiple-choice conceptual questions with distracters (incorrect responses) that reflect common student misconceptions
Think-pair-share	Posing a problem or question, having students work on it individually for a short time, and then forming pairs and reconciling their solutions. After that, calling on students to share their responses
Problem-based learning	Acting primarily as a facilitator and placing students in self-directed teams to solve open-ended problems that require significant learning of new course material
Collaborative learning	Asking students to work together in small groups toward a common goal
Cooperative learning	A structured form of group work where students pursue common goals while being assessed individually

case-based teaching can improve faculty and student attitudes, class attendance, and faculty perceptions of learning outcomes. Newson and Delatte (2011) provided example cases and advocated for their use in civil engineering courses to help students meet ABET Criterion 3 learning objectives.

Service learning Oakes (2009) defined service learning as “the intentional integration of service experiences into academic courses to enhance the learning of the core content and to give students broader learning opportunities about themselves and society at large” (p. 1). Duffy, Barry, Barrington, and Heredia (2009) reported positive influences of service learning on (1) recruitment in engineering, especially among Hispanic students, (2) student self-reports of their motivation, (3) subject matter comprehension, and (4) student self-reports of their abilities in teamwork and communication. Jiusto and DiBiasio (2006) reported positive effects on lifelong learning in the Global Studies Program at Worcester Polytechnic Institute. Dukhan, Schumack, and Daniels (2009) successfully used service learning in their heat transfer course to increase students’ social awareness and ability to identify the human connections in a project.

Think-aloud-paired problem solving Think-aloud-paired problem solving (TAPPS) is an active learning technique that can be utilized in the classroom (Lochhead & Whimbey, 1987). For TAPPS, the instructor forms teams in which one student works through a problem while the other questions the problem solver in an attempt to get students to clarify their thinking (Felder & Brent, 2009).

Inquiry learning Inquiry learning is an instructional strategy in which students are presented with questions to be answered, problems to be solved, or a set of observations to be explained (Kolb, 1984). In this study we use the term *inquiry learning* to refer to instruction that uses questions and problems to provide contexts for learning and does not fall into another more restrictive RBIS. Shymansky, Hedges, and Woodworth (1990) analyzed results from 81 experimental studies involving thousands of students and found that inquiry learning produced significant positive gains for academic achievement, student perceptions, process skills, and analytic abilities. In a meta-analysis of 79 individual studies between 1965 and 1995 involving students from seventh grade through college, Smith (1996) found that inquiry learning improved academic achievement, critical thinking skills, and laboratory skills. In a metastudy of laboratory instruction conducted over roughly the same time period, Rubin (1996) found that inquiry-based instruction was superior to traditional instruction for cognitive learning outcomes, which included conceptual and subject learning, reasoning ability, and creativity, as well as for noncognitive outcomes, including manipulative skills and attitudes. Collier (2008) used inquiry learning in his statics course and found that the electrical engineering students taking the course greatly benefited (as evidenced by higher exam grades) from hands-on inquiry learning when compared with another section which did not use inquiry learning.

Peer instruction In peer instruction (Mazur, 1997), the instructor asks students to respond to conceptual questions (often multiple choice). It is a specific case of concept tests (below) that specifies how students discuss and respond to conceptual questions. While peer instruction can be accomplished without the aid of classroom technology, it is most typically employed using classroom response systems or “clickers” (Mazur, 2009). Crouch and Mazur (2001) examined 10 years of data with physics students at Harvard comparing peer instruction to traditional lectures. They report that students taught using peer instruction had twice the normalized gains on the Force Concept Inventory of students in a control group. Students’ quantitative problem-solving skills were also measured using the Mechanics Baseline Test and results showed that peer instruction resulted in problem-solving skills that were at least equal to and generally better than those for traditionally taught courses. Additional support for the benefits of peer instruction for both promoting conceptual understanding and problem-solving skills in calculus is provided by Pilzer (2001). Koretsky and Brooks (2011) used peer instruction in a thermodynamics course for chemical engineers and found that the quick feedback provided to the faculty greatly aided them in helping students address misconceptions. On simple questions, they found an increase in student knowledge and understanding after completing the peer instruction exercise.

Concept tests Concept tests are an example of active learning where the instructor uses multiple-choice question about a course-related concept, often with distracters (incorrect responses) that reflect common student misconceptions (Felder & Brent, 2009). Santi (2007) used concept tests in a geological engineering course to address two common misconceptions.

Think-pair-share Think-pair-share is a quick active learning strategy where the instructor poses a problem, has students work on it individually for a short time, and then has them form pairs and reconcile their solutions before calling on them to share their responses

(Felder & Brent, 2009). In a thermodynamics course, Byerley (2001) used multiple active learning techniques, including think-pair-share, and found an increase in student engagement and project performance.

Problem-based learning Problem-based learning (PBL) begins by posing an open-ended, authentic problem that provides the context and motivation for learning and requires students to gather information for its solution. Most PBL implementations involve students working in self-directed teams to identify learning needs and develop viable solutions to the problems, with instructors acting as facilitators rather than primary sources of information (Prince & Felder, 2007). A meta-analysis of the effectiveness of problem-based learning (Dochy, Segers, Van den Bossche, & Gijbels, 2003) suggests that students may acquire more knowledge in the short term when taught conventionally but are likely to retain knowledge longer when taught with problem-based learning. The results for skill development consistently favored PBL instruction. Prince (2004) examined several meta-analyses and concluded that PBL improves students' skill development, retention of knowledge, and ability to apply learned material. Prince and Felder (2006) cited studies reporting a robust positive effect of PBL on development of a variety of problem-solving skills, conceptual understanding, ability to apply metacognitive and reasoning strategies, teamwork skills, and attendance. Reeves and Laffey (1999) used PBL in an Introduction to Engineering course and found an increase in the students' problem-solving skills.

Collaborative learning and cooperative learning There is little consensus in the literature and in faculty discussions regarding the difference between collaborative and cooperative learning (Barkley, Cross, & Major, 2005; Prince, 2004). For the purposes of this study, we distinguished between the two by whether the same grade is earned by all students in the group (collaborative learning) or if at least a portion of the grade is individual (cooperative learning). This definition is consistent with much of the literature in engineering education, which tends to define cooperative learning as a structured form of group work that includes, at minimum, some element of individual accountability. Effectively, this definition makes cooperative learning a special case of the more general collaborative learning. We note that this distinction is not universally reflected in the literature. For example, Barkley et al. (2005) focus on the role of the instructor and the existence of a known solution in making their distinction.

Collaborative learning, in this study, refers to any instructional method in which students work together in small groups toward a common goal (McNamee, Roberts, & Williams, n.d.). As such, it is a catch-all category of team- or group-based strategies that do not fall into other better-defined categories. (Studies traditionally cited in support of cooperative learning apply to collaborative learning as well.) Analyses of the educational impact of working together versus working individually are consistently positive. In a review of 90 years of research, Johnson, Johnson, and Smith (1998a) concluded that working together improved learning outcomes compared with individual work. In an updated study, Johnson, Johnson, and Smith (1998b) found similar results when they examined 168 studies between 1924 and 1997. Springer, Stanne, and Donovan (1997, 1999) found similar results from 37 studies of STEM students working in small groups. Shooter and McNeal (2002) used collaborative learning in their interdisciplinary mechatronics course to help students meet ABET Criterion 3 learning objectives.

In this study, cooperative learning is a structured form of group work where students pursue common goals while being assessed individually (Millis & Cottell, 1998; Prince, 2004). Cooperative learning has extensive empirical support for promoting a range of important learning outcomes (Johnson et al., 1998a, 1998b; Prince, 2004). Hsiung (2012)

divided a planar dynamics course into two sections, one using cooperative learning and the other using individualistic learning. Over time, there was an increased performance with the cooperative learning group on both homework and tests.

Methods

Overview

National faculty surveys were administered at two points in time. In spring 2011, we surveyed chemical engineering faculty who had recently taught sophomore introductory thermodynamics, fluid mechanics, or heat transfer and electrical and computer engineering faculty who had recently taught sophomore circuits, electronics, or introductory digital logic or digital design. On the basis of these results, the survey instrument was revised slightly. In spring 2012, we surveyed faculty who had recently taught statics (regardless of their disciplinary affiliation).

Development of Critical Components

Following Hall and Loucks (1978) and Mowbray et al. (2003), we developed critical components by consulting the literature describing each RBIS. As experts on the engineering and physics RBIS literature, we achieved consensus on a list of required and indicative elements, or critical components, for each RBIS. Required elements are activities that would be absolutely necessary to claim that the RBIS is being used. For example, having students discuss problems in pairs is required for think-pair-share. Indicative activities are generally associated with RBIS use, but are not required by the literature. For example, reporting out on the conclusions of a student group discussion is frequently indicative of cooperative learning but is not required to claim that cooperative learning is being used. The final list comprised 16 items, many of which correspond to multiple RBIS. (This mapping is presented in the results tables below; Table 15 is a complete list.)

Instrument

We adapted the survey instruments from a previous survey of introductory physics instructors (Henderson & Dancy, 2009; Henderson, Dancy, & Niewiadomska-Bugaj, 2012). The engineering instrument was divided into three sections. The first asked faculty to estimate the amount of class time spent on different activities generally associated with RBIS use – the required and indicative critical components. The second asked faculty about their level of use and knowledge of the 11 specific RBIS; the descriptions in Table 2 were included. The third section included demographic information such as gender, rank, and frequency of attendance at teaching workshops. To measure the reliability of the survey, Cronbach's alpha was calculated for the 2011 survey as 0.8, an acceptable level of reliability (Pedhazur & Schmelkin, 1991). On the basis of the results of the 2011 data collection, three RBIS (case-based teaching, think-aloud-paired problem solving, and service learning) were removed from the survey due to low reliability or levels of use. Three critical components which did not correspond to any remaining RBIS were also removed. The 2012 instrument was calculated to have a Cronbach's alpha of 0.9, an acceptable level of reliability (Pedhazur & Schmelkin, 1991).

Data Collection

Instructors of the courses of interest were identified by directly contacting each department.

We compiled lists of all the ABET accredited programs in chemical, electrical, and computer engineering the United States. To identify statics instructors, we compiled a list of all accredited U.S. mechanical engineering programs ($n = 285$) as well as seven civil engineering and four aeronautical/aerospace engineering programs at institutions that do not offer mechanical engineering. Virginia Tech Center for Survey Research (CSR) staff contacted each department by telephone with e-mail follow-up to identify the instructors of the courses of interest. (The protocol for statics included identifying which department was responsible for the course and following up as appropriate.)

Each instructor was invited to complete the survey by means of a personalized e-mail signed by either Margot Vigeant of the American Institute of Chemical Engineers (AIChE) survey committee, Susan Lord, then-president of the Institute for Electrical and Electronics Engineers (IEEE) Education Society, or Paul Steif and Anna Dollár, established statics professors and researchers. Gift cards were also offered as raffle incentives to those who completed the survey. The surveys of chemical, electrical, and computer engineering faculty were administered in fall 2011, and the survey of statics faculty was administered in fall 2012. To understand potential survey bias and increase the response rates, CSR contacted initial non-respondents (chemical, electrical, and computer engineering) in spring 2012 and requested they complete the survey. The early and late responses were compared, and no statistically significant differences were found between demographic information, RBIS use, or time spent on activities in class. The responses were combined into one dataset. Similarly, comparisons between the responses of faculty from different engineering disciplines yielded few statistically significant differences (Prince, Borrego, Henderson, Cutler, & Froyd, in press). Because there were few differences by discipline or course of respondent, and because three individual datasets would be too small to allow for comparisons of users and nonusers (warranting heavier reliance on Fisher's exact test), the datasets were combined into one.

Responses were screened to ensure respondents had taught one of the courses of interest and completed a majority of the survey items. Participants who did not meet these characteristics were removed from the analysis, leaving 99 chemical engineers, 122 electrical or computer engineers, and 166 statics faculty with usable responses.

The chemical engineering and electrical and computer engineering surveys were sent to 1425 faculty with 221 responses for an overall response rate (including late respondents) of 16%. The statics survey was sent to 764 faculty members with 166 responses and a response rate of 22%. Overall, we had 387 respondents, and a response rate of 18%.

Of the 387 usable responses, 19% of respondents were female and 73% male (8.2% did not respond); 9.8% were lecturers (i.e., not tenure track), 24% assistant professors, 67% associate professors, 26% full professors, and 5.9% listed their position as other (7.7% did not respond). The respondents came from a variety of engineering departments or programs: 32% electrical or computer engineering, 26% chemical engineering, 14% mechanical engineering, 15% civil, 2.1% aerospace/aeronautical, 1.0% engineering mechanics, and 2.8% indicated other (7.7% did not respond).

Data Analysis

Statistical analysis was conducted using IBM SPSS Statistics 21.0 software, released August 2012. We operationalized fidelity as the percentage of RBIS users who also spent class time on the required critical components. We identified from one to four required critical

components for each RBIS. In the Results section, we report the percentage of RBIS users who spent time on four, three, two, one, or no required components (as relevant).

Both types of survey items were multiple choice. The options for RBIS included just one option for current use; all other responses were considered to indicate nonusers. The prompt for critical components was "Please indicate what percentage of time on average your students spent/spend on each of the activities below during class time." The options were 0%, 1% to 25%, 26% to 50%, 51% to 75%, and 76% to 100%. While we can argue that problem-based learning and inquiry learning require a substantial proportion of class time, the literature provides little guidance as to the appropriate percentages we might employ in this study. Additionally, the overwhelming majority of nonzero responses were 1% to 25%. Thus all of the nonzero responses were combined to differentiate respondents who spent time on the activity from respondents who did not.

Since all respondents answered questions about which classroom activities they spent time on, we can also use this dataset to compare users to nonusers and determine whether various critical components are useful for discriminating between RBIS users and nonusers. We used chi square or Fisher's exact tests to examine the relationships between RBIS use and the classroom activities identified as critical components. (Unless indicated, all results are chi square.) The chi square test tests the null hypothesis: for example, there is no relationship between collaborative learning and having students work on a problem in pairs or groups. If the p -value is less than alpha (0.01 due to the large number of comparisons being made), the null hypothesis is rejected, which indicates a relationship between the RBIS and the activity. In other words, when an activity is required or indicative for a specific RBIS, we would expect a p -value less than 0.01 to indicate discrimination between users and nonusers. (It is also possible that both users and nonusers have a similar level of use that is not statistically significantly different. That level, if high, would indicate high fidelity of implementation without strong discrimination between RBIS users and nonusers.)

For each of these significance tests, we also calculated the effect size using phi, the appropriate test for our binary data. The effect size allows the chi square or Fisher's exact test result to be evaluated with respect to the number of tests that are run. It is generally accepted that an effect size below 0.1 indicates there is no significant relationship between the variables. Between 0.1 and 0.3, there is a slight relationship between the variables. Between 0.3 and 0.5 represents a moderate relationship, and above 0.5 is a strong relationship (Morgan, Leech, Gloeckner, & Barrett, 2011).

Limitations

As discussed in our other publications (Prince et al., in press), the limitations of this faculty survey approach include response bias from particularly conscientious instructors and self-reports of RBIS use. Both would tend to overestimate the level of RBIS use (e.g., Ebert-May et al., 2011). The goal of this analysis, though, is not to determine the proportion of engineering faculty members using RBIS; readers are cautioned against using the data for this purpose. Rather, this analysis focuses on engineering faculty members' understanding and adaptation of the RBIS, many of which were not developed in engineering. Additional responses from faculty who do not use RBIS, though likely more representative of the population, would have been of little use here. By triangulating two types of survey items against each other, we can better describe the inherent limitations to faculty surveys of teaching, which are unlikely to go away any time soon, given the relative ease with which surveys generate large amounts of quantitative data.

In general, it is difficult to estimate the degree to which response bias has skewed the results of any survey, because survey analysts, in general, do not have data on the individuals who did not respond. In surveys about health care, for example, survey analysts often have an alternate source of data, i.e., health insurance databases (Etter & Perneger, 1997). We are unaware of an alternate database that we could use for nonrespondents in our study. In fact, we first had to create our database of faculty teaching core courses, because there was no existing database to use.

Given this constraint, several precautions were taken to characterize and reduce response bias. First, we followed established practices for increasing response rates and working with professionals (Sudman, 1985). Without an alternate database to estimate response bias, a common method of estimating likelihood of response bias influencing results is to compare characteristics of the survey respondents to the general population and, if they are similar, conclude that the effects of response bias are reduced (Groves, 2006). Since the demographics of the survey respondents in this study are similar to the demographics of engineering faculty (Froyd, Borrego, Cutler, Henderson, & Prince, in press; Prince et al., in press), there is some confidence in the value of the survey results. Third, following established practice, the survey team contacted additional faculty members who did not respond to the first rounds of surveys as a way to compare early and late responders to estimate effects of response bias (Ferber, 1948). As described above, no differences were found between early and late responders that would indicate that response bias influenced results.

Finally, this analysis assumes that if respondents reported using an RBIS and one of its critical components, then the activities are linked. The high number of RBIS and classroom activities under investigation draws attention to the possibility of confounding factors that may tend to overestimate both fidelity of implementation and the discrimination of critical components. We note that the results are well structured in the sense that they reflect expected relationships consistent with literature definitions of the RBIS. Again, readers are cautioned against taking these results to represent the percentage of the engineering faculty using RBIS.

Results

Table 3 presents summary results for the fidelity of engineering science faculty's use of RBIS, specifically, the percentage of self-reported RBIS users who spent class time on four, three, two, one, or no required critical components for each RBIS. (In Table 3, *na* designates RBIS with fewer than four critical components.) The highest levels of fidelity were measured for case-based teaching (80%), just-in-time teaching (74%), concept tests (73%), and inquiry learning (70%), which all had only one required critical component. More moderate levels were measured for think-aloud-paired problem solving (62%, two required components), cooperative learning (57%, three components), collaborative learning (53%, three components), problem-based learning (52%, four components), and think-pair-share (52%, three components). Lowest fidelity was measured for peer instruction (only 11% of users spent time on all four required components) and service learning (50% of users spent time on the one required component).

It is difficult to find benchmarking fidelity scores in the literature, since many are based on rubrics developed by the researchers to study a specific program or intervention. These rubrics are not typically normed to 100% (e.g., Blakely et al., 1987; Mills & Ragan, 2000). Macias, Propst, Rodican, and Boyd (2001) reported percentages for their fidelity study of a specific model of mental health community-based care. They reported that 95%

Table 3 Percentage of Users Who Spent Time on Required Critical Components for each Research-Based Instructional Strategies (RBIS)

RBIS	Number of required critical components				
	4	3	2	1	0
Just-in-time teaching	na	na	na	74%	26%
Case-based teaching	na	na	na	80%	20%
Service learning	na	na	na	50%	50%
Think-aloud-paired problem solving	na	na	62%	29%	10%
Inquiry learning	na	na	na	70%	30%
Peer instruction ^a	11%	45%	37%	5%	2%
Concept tests	na	na	na	73%	27%
Think-pair-share	na	52%	36%	11%	1%
Problem-based learning	52%	24%	18%	5%	2%
Collaborative learning	na	53%	35%	10%	2%
Cooperative learning	na	57%	36%	6%	2%

^aIn statics survey, one critical component was separated into two items (uses clickers and uses other means)
na = an RBIS had fewer than four critical components

of all sites reported following at least half of the required practices. By this standard, our fidelity results are quite low. (For example, Table 3 shows 18% of PBL users spent time on two of four required critical components, as compared with 95% in the mental health study.) Lack of results with which to benchmark is a limitation of this study as well as an opportunity for future work. To explore these results in greater detail, we discuss the critical and required components for each RBIS in more detail below.

In general, the percentage of RBIS users who spent class time on each critical component is much higher than the combined percentages for multiple components reported in Table 3. The RBIS-level tables are more appropriate for addressing the second research question of which critical components discriminate between RBIS users and nonusers. Significance tests for every combination of RBIS and classroom activity were run, regardless of whether they were designated critical components. Tables 3 to 14 report the results for every required and indicative component. Unexpected significant relationships are reported in the text only.

Just-in-Time Teaching

The results for the just-in-time teaching RBIS are presented in Table 4. JiTT demonstrated relatively high fidelity with 74% users and 49% of nonusers spending time on the required activity (setting deadlines before class and adjusting their lecture). This activity also discriminates between users and nonusers ($p = 0.001$, $\phi = 0.163$ indicates a slight relationship). This discrimination is to be expected since it is such a specific activity to JiTT. The indicative activity of engaging students also showed a majority (76%) of users also spending time on the activity but did not discriminate. (The engaging students activity was actually mapped to all 11 RBIS as indicative, so it would be surprising if it was statistically significant here.)

An additional significant relationship was found between JiTT and having students “participate in group work for which the assessments are designed so that individuals may earn different scores for their work on the assignment” ($p = 0.001$). Half (50%) of faculty who use JiTT spent time on this activity, while only 26% of nonusers spent time. It is

Table 4 Just-in-Time Teaching Required and Indicative Components

Student activity	Type	Faculty who use RBIS (<i>n</i> = 46) and spent time on activity	Faculty who don't use RBIS (<i>n</i> = 341) but spent time on activity	<i>p</i> -value	phi
Spent time discussing pre-class activities which helped you re-evaluate student learning and adjust your lecture 'just in time'	Required	74%	49%	0.001 ^a	0.163
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening, and/or taking notes)	Indicative	76%	67%	0.222	0.062

^aIndicates significance with alpha = 0.01

possible that this is an adaptation of JiTT that is worth further investigation, or JiTT users may simply tend to also assign group work in their classes.

Case-Based Teaching

Similarly, case-based teaching demonstrated high fidelity and discrimination for its one, unique required critical component. As listed in Table 5, fully 80% of users but only 36% of nonusers required students to analyze case studies, a statistically significant difference. For the indicative components, RBIS users ranged from 60% to 88%, which demonstrates relatively high fidelity but not discrimination in most cases. Again, many of these activities overlap with other RBIS, and nonusers also reported participating at high levels (43% to 70%).

An additional significant relationship was found between case-based teaching and having students “participate in group work for which assessments are designed so that individuals may earn different scores for their work on the assignment” (*p* = 0.008). About half (52%) of RBIS users spent time on this activity, but less than a third (27%) of nonusers spent time on the activity. It is reasonable to consider in future work that an adaptation of case-based teaching in engineering may be to have students discuss cases in groups but have students responsible for their own written analysis of the case.

Service Learning

Although service learning is gaining acceptance in engineering education, we did not expect to see it used extensively in engineering science courses. In fact, only 12 respondents said they currently use it. As indicated in Table 6, only 50% of users reported spending time on the one required critical component, assigning projects with real clients in the university or local community. The relatively low 50% may be accounted for in part by the changing nature of service learning, which has an increasingly global focus. Nonetheless, this critical component effectively discriminated between users and nonusers (*p* = 0.001, phi = 0.311 indicates a moderate relationship). Finally, the indicative activity, although not statistically significant, demonstrates high fidelity with 92% of users engaging students with course content.

Think-Aloud-Paired Problem Solving

The results for think-aloud-paired problem solving (TAPPS) are presented in Table 7. The three critical components demonstrate relatively high fidelity among users (71% to

Table 5 Case-Based Teaching Required and Indicative Components from Electrical and Computer Engineering and Chemical Engineering Surveys

Student activity	Type	Faculty who use RBIS ($n = 25$) and spent time on activity	Faculty who don't use RBIS ($n = 196$) but spent time on activity	p -value	phi
Analyze case studies of historical or hypothetical situations that involve solving problem or making decisions	Required	80%	36%	$< 0.001^{a,b}$	0.282
Work on projects inspired by problems or situations from real engineering	Indicative	88%	70%	0.062 ^a	0.128
Complete specifically designed activities to 'learn' course concepts on their own without being explicitly told	Indicative	60%	43%	0.116	0.106
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	72%	62%	0.317	0.067
Discuss a problem in pairs or groups	Indicative	72%	70%	0.829	0.015

^aFisher's exact test used due to less than $n = 5$ in a cell^bIndicates significance with $\alpha = 0.01$ **Table 6** Service Learning Required and Indicative Components from Electrical and Computer Engineering and Chemical Engineering Surveys

Student activity	Type	Faculty who use RBIS ($n = 12$) and spent time on activity	Faculty who don't use RBIS ($n = 209$) but spent time on activity	p -value	phi
Work on a project with a real, not-for-profit client in the university of local community	Required	50%	8.1%	$< 0.001^b$	0.311
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	92%	61%	0.35 ^a	0.143

^aFisher's exact test used due to less than $n = 5$ in a cell^bIndicates significance with $\alpha = 0.01$

81%). These results, though, show no significant relations between the three critical components and TAPPS. The lack of a significant relationship is not particularly surprising, since each of these components is shared with multiple other RBIS.

An additional significant relationship was found between TAPPS and "time discussing pre-class activities which helped you re-evaluate student learning and adjust your lecture

‘just in time’” ($p = 0.01$). Nearly two-thirds (71%) of TAPPS users also spent time on this activity, but less than half (42%) of nonusers spent time. This relationship may represent an engineering adaptation to TAPPS, or TAPPS users may simply also tend to employ some aspects of JiTT. This is another possibility for future work to explore.

Inquiry Learning

The results for inquiry learning (Table 8) indicate high fidelity with significant relationships between the RBIS and all three of its critical components. Inquiry learning users assign problems or projects that require students to seek out new information (70% of users, $p < 0.001$, $\phi = 0.178$ indicates a slight relationship), design activities for students to learn material without being told (66% of users, $p < 0.001$, $\phi = .220$ indicates a slight relationship), and generally engage students with the material (84% of users, $p < 0.001$, $\phi = .231$ indicates a slight relationship).

Three additional statistically significant results were found for this RBIS. First, inquiry learning users are more likely (54%) than nonusers (37%) to have students “report their group’s findings to the entire class (formally or informally)” ($p = 0.001$). It is reasonable to consider this as a potential adaptation of inquiry learning to investigate in future work.

Second, more inquiry learning users (86%) than nonusers (71%) have students “work on projects inspired by problems or situations from real engineering practice” ($p = 0.001$). This is another potential adaptation to consider that may be closely aligned to the practical and professional nature of engineering (in contrast with other science education settings where inquiry learning was first developed).

Third, inquiry learning users (65%) were more likely than nonusers (45%) to have their students spend “time discussing pre-class activities which helped you re-evaluate student learning and adjust your lecture ‘just in time’” ($p < 0.001$). This is yet another potential adaptation to explore in future work, or it may be a correlation between inquiry learning and JiTT users.

Table 7 Think-Aloud-Paired Problem Solving Required and Indicative Components from Electrical and Computer Engineering and Chemical Engineering Surveys

Student activity	Type	Faculty who use RBIS ($n = 21$) and spent time on activity	Faculty who don’t use RBIS ($n = 200$) but spent time on activity	p -value	ϕ
Explain their reasoning to another student while solving a specific problem	Required	71%	54%	0.116	0.106
Discuss a problem in pairs or groups	Required	81%	69%	0.322 ^a	0.077
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	76%	62%	0.238 ^a	0.089

^aFisher’s exact test used due to less than $n = 5$ in a cell

Table 8 Inquiry Learning Required and Indicative Components

Student activity	Type	Faculty who use RBIS ($n = 124$) and spent time on activity	Faculty who don't use RBIS ($n = 263$) but spent time on activity	p -value	phi
Work on problems or projects that require students to seek out new information not previously covered in class	Required	70%	51%	$< 0.001^a$	0.178
Complete specially designed activities to 'learn' course concepts on their own without being explicitly told	Indicative	66%	42%	$< 0.001^a$	0.220
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening, and/or taking notes)	Indicative	84%	61%	$< 0.001^a$	0.231

^aIndicates significance with $\alpha = 0.01$

Peer Instruction and Concept Tests

Peer instruction and concept tests (or ConcepTests) are closely related RBIS. In fact, many STEM undergraduate education Web sites credit both to Eric Mazur. Consistent with the literature, we mapped concept tests to the conceptual questions and peer instruction to the activities of posing conceptual questions *and* having students discuss and report on their answers. (These required components of peer instruction are indicative components for concept tests.)

The results in Tables 9 and 10 demonstrate many similarities in the responses related to concept tests and peer instruction users, with slightly stronger evidence for the additional required components for peer instruction. For both RBIS, multiple-choice conceptual questions were relatively high and statistically significant (71% of peer instruction users vs. 35% of nonusers, $p < 0.001$, $\phi = 0.265$ indicates slight relationship; 73% of concept test users vs. 31% of nonusers, $p < 0.001$, $\phi = .366$ indicates moderate relationship). Discussing a problem in pairs or groups demonstrated high fidelity (92% of peer instruction users and 81% of concept tests users) for both, but discriminated only between peer instruction users and nonusers ($p = 0.004$). Clickers or similar means for students to report their answers discriminated between peer instruction and concept tests users ($p < 0.001$ for both, moderate relationship for peer instruction), but fidelity was not as high (61% peer instruction users and 48% concept tests users). Requiring students to provide an answer before class can proceed had high fidelity for both RBIS (92% of peer instruction users, 81% of concept test users) but did not discriminate between users and nonusers. Reporting back to the class was designated as an indicative activity for only peer instruction, with moderate fidelity (58% of users) that nonetheless discriminated between users and nonusers ($p = 0.006$). Finally, engaging students in course content demonstrated high fidelity (81% of peer instruction users and 76% of concept test users) but did not discriminate. In summary, engineering science faculty report using concept tests and peer instruction in similar ways, with perhaps some preference for clickers, discussion, and report-back in peer instruction, which is consistent with the literature. Since peer instruction

Table 9 Peer Instruction Required and Indicative Components

Student activity	Type	Faculty who use RBIS ($n = 62$) and spent time on activity	Faculty who don't use RBIS ($n = 325$) but spent time on activity	p -value	phi
Answer multiple-choice conceptual questions with distracters (incorrect responses) that reflect common student misconceptions	Required	71%	35%	$< 0.001^a$	0.265
Discuss a problem in pairs or groups	Required	92%	76%	0.004 ^a	0.145
Use clickers or similar means to 'vote' on the correct answer of a multiple choice question ^b	Required	61%	22%	$< 0.001^a$	0.318
Provide answer(s) to a posed problem or question before the class can proceed	Required	84%	77%	0.245	0.059
Report their group's findings to the entire class (formally or informally)	Indicative	58%	23%	0.006 ^a	0.141
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	81%	66%	0.022	0.117

^aIndicates significance with $\alpha = 0.01$ ^bIn statics survey, one critical component was separated into two items (uses clickers and uses other means)

had four required critical components and concept tests only one, the overall fidelity of implementation of peer instruction (11%, Table 3) is much lower than concept tests (73%).

Beyond the critical components mapped to peer instruction, additional and unexpected statistically significant relationships suggest that engineering faculty are either adapting or misinterpreting the concept of peer instruction. Specifically, there were differences for having students "complete specially designed activities to 'learn' course concepts on their own without being explicitly told" (69% of peer instruction users, 47% of nonusers, $p = 0.001$) and having students "work on projects inspired by problems or situations from real engineering practice" (94% of users, 73% of nonusers, Fisher's exact test $p < 0.001$).

On the basis of the nature of these activities, namely their incompatibility with quick conceptual questions discussed by students during a lecture, we consider the possibility that some respondents are misinterpreting the term *peer instruction* to refer to tutoring or other group work. (Our analysis of transcripts of interviews with engineering science faculty suggests that one quarter to one third of faculty interpret peer instruction in this way, even when they are discussing a handout with Mazur's definition as part of the interview protocol.) It is also possible that further investigation would conclude that faculty who use peer instruction are also more likely to assign projects or inquiry learning experiences. Yet the fact that these were not significant for concept tests reinforces that it is a phenomenon unique to peer instruction, perhaps due to its potentially misleading name.

Table 10 Concept Test Required and Indicative Components

Student activity	Type	Faculty who use RBIS ($n = 95$) and spent time on activity	Faculty who don't use RBIS ($n = 292$) but spent time on activity	p -value	phi
Answer multiple-choice conceptual questions with distracters (incorrect responses) that reflect common student misconceptions	Required	73%	31%	$< 0.001^a$	0.366
Discuss a problem in pairs or groups	Indicative	81%	77%	0.453	0.038
Use clickers or similar means to 'vote' on the correct answer of a multiple choice question ^b	Indicative	48%	22%	$< 0.001^a$	0.253
Provide answer(s) to a posed problem or question before the class can proceed	Indicative	79%	78%	0.859	0.009
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	76%	66%	0.068	0.093

^aIndicates significance with $\alpha = 0.01$ ^bIn statics survey, one critical component was separated into two items (uses clickers and uses other means)

Think-Pair-Share

The results for the individual critical components of think-pair-share (TPS), given in Table 11, demonstrate relatively high fidelity, even though the combined level of fidelity is 52% (Table 3). TPS users report spending class time on discussing problems in pairs or groups (94%), reporting the findings to the class (64%), and waiting for these answers before proceeding (82%). Users report otherwise engaging students with course content (indicative, 86%). Most of these critical components discriminated between users and nonusers ($p < 0.001$), with the exception of waiting for student answers to proceed with class. Again, this item was shared by several RBIS and used by 78% of all respondents but was not significant for any RBIS.

Four additional statistically significant relationships were found between TPS use and classroom activities. Each is related to the principles of TPS and might be considered either as a potential adaptation to be explored in future research or as a close relationship between TPS and other similar RBIS. First, more TPS users (63%) than nonusers (46%) have their students "complete specially designed activities to 'learn' course concepts on their own without being explicitly told" ($p = 0.005$). Second, 81% of TPS users and 67% of nonusers ask their students to "work on problem sets or projects in pairs or small groups" ($p < 0.001$). Third, almost twice as many TPS users (43%) as nonusers (24%) "use clickers or similar means to 'vote' on the correct answer of multiple-choice question" ($p < 0.001$). Fourth, 53% of TPS users and 37% of nonusers ask their students to "answer multiple-choice conceptual questions with distracters (incorrect responses) that reflect common student misconceptions" ($p = 0.007$).

Table 11 Think-Pair-Share Required and Indicative Components

Student activity	Type	Faculty who use RBIS (<i>n</i> = 92) and spent time on activity	Faculty who don't use RBIS (<i>n</i> = 295) but spent time on activity	<i>p</i> -value	phi
Discuss a problem in pairs or groups	Required	94%	65%	< 0.001 ^a	0.206
Report their group's findings to the entire class (formally or informally)	Required	64%	35%	< 0.001 ^a	0.249
Provide the answer(s) to a posed problem or question before the class session can proceed	Required	82%	77%	0.390	0.044
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	86%	63%	< 0.001 ^a	0.212

^aIndicates significance with alpha = 0.01

Problem-Based Learning

Although overall fidelity of implementation for problem-based learning (PBL) was moderate (52%, Table 3), the fidelity of the individual required components was much higher, ranging from 72% to 85% of users (Table 12). Three of four required activities discriminated between users and nonusers (learning course concepts on own, working in groups, and problems inspired by engineering practice). All three indicative components were statistically significant ($p < 0.001$). The two indicative activities with the lowest fidelity (44% and 58%) are grading schemes (same or different project grades for group members) that may be mutually exclusive for many faculty members. Thus, a large percentage of PBL users also have students participate in group work but vary in how the group work is graded.

Collaborative and Cooperative Learning

As demonstrated by similarities between Tables 13 and 14, collaborative and cooperative learning share most of their critical components. For this study, we distinguished between the two by whether the same grade is earned by all students in the group (collaborative learning) or if at least a portion of the grade is individual (cooperative learning). (This distinction is not universally reflected in the literature.) Cooperative learning, though, includes reporting back to the class as an indicative component.

Overall, the results are very positive: 60% to 96% of users spent time on required and indicative components, and there were statistically significant differences between users and nonusers ($p < 0.001$ for nine of 10 tests, all with slight or moderate effect sizes). Again, waiting for students to provide the answer before class can proceed was not statistically significant for any RBIS, but it was practiced by 80% to 84% of collaborative and cooperative learning users.

The grading distinction between cooperative and collaborative learning was not reflected in the results. Both collaborative and cooperative learning had statistically significant relationships

Table 12 Problem-Based Learning Required and Indicative Components

Student activity	Type	Faculty who use RBIS ($n = 124$) and spent time on activity	Faculty who don't use RBIS ($n = 263$) but spent time on activity	p -value	phi
Complete specifically designed activities to 'learn' course concepts on their own without being explicitly told	Required	72%	42%	$< 0.001^a$	0.262
Work on problem sets or projects in pairs or small groups	Required	78%	50%	0.001 ^a	0.175
Work on projects inspired by problems or situations from real engineering practice	Required	85%	73%	0.009 ^a	0.133
Discuss a problem in pairs or groups	Required	84%	76%	0.076	0.090
Participate in group work for which they earn the same score as every other member of the group	Indicative	58%	37%	$< 0.001^a$	0.187
Participate in group work for which the assessments are designed so that individuals may earn different scores for their work on the assignment	Indicative	44%	23%	$< 0.001^a$	0.200
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	84%	62%	$< 0.001^a$	0.210

^aIndicates significance with $\alpha = 0.01$

with both types of grading (same grade for all group members vs. different grades). This relationship can be explained in part because instructors who assign group work are more likely to assign group grades than are instructors who assign individual work. Yet more users of cooperative learning assign group grades (66%), which we mapped to collaborative learning, than individual grades (61%), which we mapped to cooperative learning. This distinction, as well as possible overlaps in the data, should be explored further and reconsidered in future research.

Both RBIS were also found to have significant relationships with additional activities including working on problems inspired by real engineering practice ($p < 0.001$ for collaborative learning; $p < 0.001$ using Fisher's exact test for cooperative learning) and having students seek out new information ($p < 0.001$ for collaborative learning; $p < 0.001$ for cooperative learning). This result may help to characterize the types of group work assigned in engineering science courses, that is, providing evidence that engineering group problems or projects are often based on real engineering practice and require students to seek out new information.

Cooperative learning also presented an unexpected significant relationship with spending "time discussing pre-class activities which helped you re-evaluate student learning and adjust your lecture 'just in time'" ($p = 0.002$). Although this activity was not significantly related to

Table 13 Collaborative Learning Required and Indicative Components

Student activity	Type	Faculty who use RBIS (<i>n</i> = 205) and spent time on activity	Faculty who don't use RBIS (<i>n</i> = 182) but spent time on activity	<i>p</i> -value	phi
Discuss a problem in pairs or groups	Required	91%	64%	< 0.001 ^a	0.320
Work on problem sets or projects in pairs or small groups	Required	88%	53%	< 0.001 ^a	0.387
Participate in group work for which they earn the same score as every other member of the group	Required	60%	24%	< 0.001 ^a	0.367
Complete specially designed activities to 'learn' course concepts on their own without being explicitly told	Indicative	60%	39%	< 0.001 ^a	0.210
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	82%	53%	< 0.001 ^a	0.313
Provide the answer(s) to a posed problem or question before the class session can proceed	Indicative	80%	77%	0.537	0.031

^aIndicates significance with alpha = 0.01

collaborative learning, the relationship between the two may stem from having students work cooperatively outside of class on an assignment and then individually turn in and receive an individual grade for the assignment. This potential adaptation should be investigated further.

Discussion and Future Work

This study used a faculty survey to explore fidelity of implementation of 11 RBIS in engineering science courses. Fidelity was quantified as the percentage of RBIS users who also spent class time on the required critical components. As indicated by the results in Table 3, overall fidelity is strongly influenced by the number of components. Four of five RBIS with one required critical component earned overall fidelity scores of 70% or higher (case-based teaching, just-in-time teaching, concept tests, and inquiry learning). In contrast, many other RBIS with three or four required components scored 50% to 60% fidelity of implementation (cooperative learning, collaborative learning, problem-based learning, and think-pair-share). Many of the individual required components demonstrated much higher fidelity, above 70% and as high as 96%. Service learning demonstrated the lowest fidelity (50% of users spent time on the one critical component).

There is very little prior work by which to judge whether these values are favorable. For the purposes of discussion, we used arbitrary qualifiers “relatively high” for values of 70% to 100%, “moderate” for 50% to 69%, and “low” for values below 50%. Examples from the mental health literature suggest that these are not particularly strong, and self-reporting and response bias by enthusiastic instructors are both likely to overestimate RBIS use and time spent on various activities. This study is an important first step, but clearly more work is

Table 14 Cooperative Learning Required and Indicative Components

Student activity	Type	Faculty who use RBIS ($n = 67$) and spent time on activity	Faculty who don't use RBIS ($n = 320$) but spent time on activity	p -value	phi
Discuss a problem in pairs or groups	Required	96%	75%	$< 0.001^{a,b}$	0.191
Work on problem sets or projects in pairs or small groups	Required	91%	67%	$< 0.001^b$	0.200
Participate in group work for which the assessments are designed so that individuals may earn different scores for their work on the assignment	Required	61%	22%	$< 0.001^b$	0.329
Report their group's findings to the entire class (formally or informally)	Indicative	79%	34%	$< 0.001^b$	0.343
Complete specifically designed activities to 'learn' course concepts on their own without being explicitly told	Indicative	76%	45%	$< 0.001^b$	0.238
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	Indicative	82%	65%	0.007 ^b	0.136
Provide the answer(s) to a posed question before the class session can proceed	Indicative	84%	77%	0.537	0.059

^aFisher's exact test used due to less than $n = 5$ in one cell^bIndicates significance with $\alpha = 0.01$

needed to identify benchmarks for fidelity of implementation in undergraduate education. We collected more detailed information on the percentage of class time spent on each activity that could be explored further. Depending on the RBIS, the amount of time spent on each activity will vary. For example, with think-pair-share a limited percentage of the overall class time should be spent utilizing the RBIS, but within that time students should discuss a problem in pairs and share their discussion with the overall class. This data and similar future research could contribute to understanding optimal times spent on various RBIS and improve discrimination between RBIS users and nonusers.

Most fidelity studies only examine users, but this study design allowed for comparison of RBIS users and nonusers. An additional level of analysis included chi square and Fisher's exact tests to determine whether each critical component could discriminate between users and nonusers. Due to the exploratory nature of this analysis and the high number of statistical tests run, a conservative value of α (0.01) was selected and effect size was reported. Effect size scaled with significance, indicating moderate relationships between variables at best, and suggesting that an even more conservative α would have been appropriate (e.g., 0.001) to focus on relationships with practical significance.

Table 15 summarizes the results of the statistical tests presented in Tables 4 to 14. A discrimination score was calculated for each critical component by dividing the number of significant relationships (based on alpha of 0.01) by the total number of RBIS for which it was a required or indicative component. Five classroom activities discriminated between users and nonusers for every RBIS to which they were mapped, as well as at least one additional RBIS: problems or projects that require students to seek out new information, group work on problem sets or projects, report back following group work, group grades for group work, and use of clickers or similar means.

It is interesting to note that although most of these are general strategies that are shared by many RBIS (particularly working in groups or pairs), they nonetheless had high discriminatory power. For this reason, these activities may be particularly important for future surveys, observations, and other measures of RBIS use. Nine of the remaining 11 also discriminated for every RBIS to which they were designated as a required component. Only two were unable to discriminate for any RBIS (provide answer to a posed problem before class can proceed, and explain reasoning to another student while working on a specific problem). Of these two, providing the answer to a problem before class proceeds was employed by 78% of all respondents, while only 53% of all respondents had students describe their reasoning to each other during class time. We might conclude that the lack of discrimination for the component of providing the answer to a problem is low because it is in such common use and shared by many RBIS. On the other hand, students describing their reasoning to each other may be nondiscriminating because it (and its corresponding RBIS, TAPP) is not used widely in engineering. These items that did not discriminate well should be considered for revision before similar studies are administered in the future.

In their study of fidelity of community mental health care programs, Macias et al. (2001) also used significance tests to discriminate between certified and noncertified programs. They found that 33 of 42 components effectively discriminated between certified and noncertified programs. Most of their items that did not discriminate were practiced by less than 50% of certified and noncertified sites. Although the overall fidelity values (percentages) achieved in our study of engineering science faculty were much lower than their example, our discrimination results were comparable if not significantly stronger (13 of 16 = 81% vs. 33 of 42 = 78%).

In addition to the procedures described in Methods, critical components can be validated by comparing the expected and actual results. Overall, we can say that the survey results reflect a pattern that is consistent with the critical components we identified. It would be inappropriate to conclude that RBIS are being used in engineering science courses as intended by the original developers. Yet the engineering faculty who responded to this survey are knowledgeable about these RBIS, and their knowledge is consistent with what we have identified from the literature.

The final column of Table 15 also lists the unexpected statistically significant relationships between RBIS and classroom activities, the details of which are presented in the corresponding Results section for each RBIS. In each case we offered some interpretation of whether these were reasonable adaptations of the RBIS as applied in engineering sciences, a potential correlation between various activities and RBIS, or, in the case of peer instruction, an alternative interpretation to the developers' original intention. These interpretations are of course merely conjecture, and complementary approaches such as interviews and observations should be used to explore these further.

More generally, future research should continue to explore many of the measurement issues identified in this study. The limitations of faculty surveys can be better understood

Table 15 Summary of Chi Square Discrimination Test Results

Student activity (Critical component)	Required		Indicative		Discrimination	Unexpected significant relationships
	No. of RBIS	No. Significant	No. of RBIS	No. Significant		
Work on problems or projects that require students to seek out new information not previously covered in class	1	1	0	0	100%	Collaborative learning, cooperative learning
Report their group's findings to the entire class (formally or informally)	1	1	3	3	100%	Inquiry learning
Work on problem sets or projects in pairs or small groups	3	3	1	1	100%	Think-pair-share
Participate in group work for which they earn the same score as every other member of the group	1	1	2	2	100%	Cooperative learning
Use clickers or similar means to 'vote' on the correct answer of a multiple choice question	1	1	2	2	100%	Think-pair-share
Complete specially designed activities to 'learn' course concepts on their own without being explicitly told	1	1	5	4	83%	Peer instruction, think-pair-share
Participate in group work for which the assessments are designed so that individuals may earn different scores for their work on the assignment	1	1	2	1	67%	Just-in-time teaching, case-based teaching, collaborative learning
Answer multiple-choice conceptual questions with distracters (incorrect responses) that reflect common student misconceptions	2	2	1	0	67%	Think-pair-share
Discuss a problem in pairs or groups	6	4	3	1	56%	None
Spent time discussing pre-class activities which helped you re-evaluate student learning and adjust your lecture 'just in time'	1	1	1	0	50%	Think-aloud paired problem solving, inquiry learning, cooperative learning
Participate in activities that engage them with course content through reflection and/or interaction with their peers (other than watching, listening and/or taking notes)	1	1	11	5	50%	None

Table 15 (continued)

Student activity (Critical component)	Required		Indicative		Discrimination	Unexpected significant relationships
	No. of RBIS	No. Significant	No. of RBIS	No. Significant		
Work on a project with a real, not-for-profit client in the university or local community	1	1	1	0	50%	None
Analyze case studies of historical or hypothetical situations that involve solving problems or making decisions	1	1	1	0	50%	None
Work on projects inspired by problems or situations from real engineering practice	1	1	2	0	33%	Inquiry learning, peer instruction, cooperative learning, collaborative learning
Provide the answer(s) to a posed problem or question before the class session can proceed	2	0	4	0	0%	None
Explain their reasoning to another student while solving a specific problem	1	0	1	0	0%	None

and addressed using qualitative methods such as classroom observations. It is unlikely that surveys will be abandoned altogether, despite the limitations identified in the DBER report (NRC, 2012), because of the large volume of quantitative data they generate with relatively little effort and the increasing focus on scaling up effective teaching practices.

Conclusion

On the basis of these results, we conclude that fidelity of implementation is a useful framework for continued study of undergraduate STEM instruction. A rich literature in K–12 education, social work, counseling psychology, and medicine provides guidance on identifying and validating critical components and measuring fidelity, using quantitative, qualitative, and mixed methods. The fidelity of implementation literature acknowledges that educational developers must balance between adaptation and faithful replication and must communicate this balance through critical components and accompanying documentation. Developers or other change agents must also regularly review and modify configurations as needed to reflect changes in use by practicing instructors over time. Although this article focuses on research, its lessons are important to those who develop and use RBIS, and we encourage other STEM faculty to use them. Specifically, developers should consider following the steps laid out by Hall and Loucks (1978) for developing a full innovation configuration to describe completely their instructional strategy. Communication to convince others to use RBIS should include not only evidence of efficacy but also details related to implementation and the underlying educational research principles; such

inclusion will discourage engineering instructors from inappropriate adaptation that would sacrifice the documented benefits. Potential users of RBIS should seek out information on implementation and underlying educational research principles to make informed decisions and adaptations in their own teaching. User communities can help in updating and distributing this information. Following these principles that are guided by a fidelity of implementation perspective can help instructors improve engineering and STEM education for a broader group of students.

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